

Piecewise PV Inversion Handbook

The Wu Fortran Pipeline with an Isolated Surface- θ Piece

For the 2025-01-08 California Blocking Case Study

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pv_inversion/

June 4, 2026

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Chapter 1

Introduction and Overview

1.1 Purpose of this Handbook

This handbook documents **piecewise potential-vorticity inversion** (PPVI) for atmospheric blocking, as implemented in `pv_inversion/`. The goal is to take the 2025-01-08 California blocking event, partition its potential-vorticity (PV) anomaly into vertical pieces, invert each piece to its own balanced circulation, and thereby quantify how the upper-tropospheric PV anomaly, the mid/low-tropospheric PV, and the *surface thermal* anomaly each contribute to the blocking flow.

The pipeline has a single numerical core: **W. Wu’s finite-difference successive-over-relaxation (SOR) solver** (`wu/fortran/pvpialln_94UV.f`, `qinvert21_94.f`, `qinvertp21_94.f`), driven from a thin Python orchestration layer (`wu/steps/`). This Fortran is the same lineage as C. Davis’s `PVPI.FOR` [1] and the modern GFDL/CSU PPVI tool; the inversion mathematics is unchanged, only the driving and post-processing are in Python.

What this handbook is about

A *single* working track — the Wu Fortran pipeline — on a **nine-level** ERA5 grid, with the PV perturbation split into **three pieces** that mirror the reference partition $\{1\}, \{2, 3\}, \{4..N_L\}$:

piece	levels K	physical content
surface	$\{1\}$	1000 hPa boundary θ (the surface heating)
lower	$\{2, 3\}$	850, 700 hPa interior PV
upper	$\{4, 5, 6, 7, 8, 9\}$	500–200 hPa interior PV + 100 hPa top θ

The key conceptual point (Chapters 2 and 3): the **surface warm- θ anomaly is a boundary condition, not an interior PV value** — by Bretherton’s equivalence it acts as a PV *sheet* at the lower boundary, and we invert it as its own “surface” piece.

Retired material

Earlier drafts described a second “Part II” spherical-harmonic (SH) Python port and several Family-B solvers (a direct spectral solve and an LGMRES/linearized-Charney solver). **These have been removed.** The SH BALNC/BALP port never converged and is no longer maintained; this handbook documents only the Wu Fortran pipeline that runs end-to-end.

1.2 The Case: 2025-01-08 California Block

The event is a strong wintertime ridge over the northeast Pacific / west coast of North America that contributed to the January 2025 California wildfire outbreak. At the analysis time (2025-01-08 00 UTC) the 500 hPa block centre is at $\approx 31.5^\circ\text{N}$, 222°E (138°W), with $Z_{500} \approx 5890$ m. All cross-sections and the 3-D structure figure (Section 5.6) are centred on this point.

1.3 Notation Cheat-Sheet

Symbol	Meaning
ψ, Φ	streamfunction ($\text{m}^2 \text{s}^{-1}$), geopotential ($\text{m}^2 \text{s}^{-2}$)
$\bar{\psi}, \bar{\Phi}$	climatological mean (basic) state
$\delta\psi = \psi - \bar{\psi}, \delta\Phi = \Phi - \bar{\Phi}$	anomalies
q	Ertel potential vorticity ($\text{PVU} = 10^{-6} \text{K m}^2 \text{kg}^{-1} \text{s}^{-1}$)
$\bar{q}, q' = q - \bar{q}$	mean and anomalous PV
θ_b	boundary potential temperature (surface / top)
$\zeta = \nabla^2 \psi$	relative vorticity
$f = 2\Omega \sin \phi, f_0 = f(45^\circ)$	Coriolis parameter, reference value
$\pi = (p/p_0)^\kappa$	Exner function ($p_0 = 10^5$ Pa, $\kappa = R/c_p$)
$\theta = T(p_0/p)^\kappa$	potential temperature
$K = 1 \dots N_L$	1-indexed Fortran level ($K=1$ surface, $K=N_L$ top)
∇^2, ∇^{-2}	horizontal Laplacian and its inverse

A more complete glossary appears in Appendix A.

1.4 Quick Tour of the Code

Component	Purpose	Reference
<code>config.py</code> / <code>wu/wu_config.yaml</code>	grid, 9 levels, pieces, solver params	Chapter 3
<code>wu/fortran/pvpialln_94UV.f</code>	Pass A/B: Ertel PV, $\bar{\psi}$, θ_b	Chapter 3
<code>wu/fortran/qinvert21_94.f</code>	Pass C: total balanced inversion (BALNC)	Chapter 3
<code>wu/fortran/qinvertp21_94.f</code>	Pass D: piecewise perturbation (BALP)	Chapter 3
<code>wu/steps/05...11/</code>	the step-by-step pipeline + plots	Chapter 5
<code>wu/steps/08/parse_and_pv.py</code>	Wu $Q \rightarrow$ SI Ertel PV ($\times 10^{-8}$)	Chapter 4

The shared front end (`shared_steps/01...04`) downloads the ERA5 event hour, builds a 30-year climatology, and writes the Fortran `.grid` input files; the Wu steps (05–11) compile and run the four passes and produce all diagnostics.

Chapter 2

Mathematical Foundations

2.1 Coordinates and Constants

The vertical coordinate is the **Exner pressure**

$$\pi \equiv \left(\frac{p}{p_0}\right)^\kappa, \quad p_0 = 10^5 \text{ Pa}, \quad \kappa = \frac{R_d}{c_p} = \frac{2}{7} \approx 0.2857 \quad (2.1)$$

with $\pi = 1$ at the surface and $\pi \rightarrow 0$ at the model top. The Wu Fortran hardcodes $\kappa = 2/7$ (this exact value is what makes the $Q \rightarrow \text{SI}$ conversion a clean constant, Chapter 4). On the $N_L = 9$ ERA5 pressure levels used here,

$$\text{PLEVS} = [1000, 850, 700, 500, 400, 300, 250, 200, 100] \text{ hPa}, \quad (2.2)$$

$$\pi_k = [1.000, 0.955, 0.903, 0.820, 0.770, 0.709, 0.673, 0.631, 0.518]. \quad (2.3)$$

Constant	Symbol	Value
Gravity (Fortran)	g	9.81 m s^{-2}
Dry-air gas constant	R_d	$287 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat at p	c_p	$1004.5 \text{ J kg}^{-1} \text{ K}^{-1}$ (Fortran)
$\kappa = R_d/c_p$	κ	$2/7 \approx 0.2857$
Earth's rotation	Ω	$7.292 \times 10^{-5} \text{ rad s}^{-1}$
Earth radius (Wu)	$a = 2 \times 10^7 / \pi$	$6.366 \times 10^6 \text{ m}$
Reference pressure	p_0	10^5 Pa
Reference Coriolis	f_0	$f(45^\circ) \approx 1.031 \times 10^{-4} \text{ s}^{-1}$

2.2 Thermodynamic Relations

Potential temperature

$$\theta = T \left(\frac{p_0}{p}\right)^\kappa. \quad (2.4)$$

The Fortran is fed *raw temperature* T and forms θ internally as $\theta = T c_p / \Pi$ with $\Pi = c_p (p/p_0)^\kappa$ (`pvpialln_94UV.f`); feeding it θ would double-convert and inflate the static stability.

Hydrostatic balance in Exner coordinates

$$\frac{\partial \Phi}{\partial \pi} = -c_p \theta \quad (2.5)$$

is the vertical link between Φ and θ that closes the inversion and *defines the boundary condition* of Section 2.4.

2.3 Ertel Potential Vorticity

The hydrostatic Ertel PV is

$$Q = -g(\zeta + f) \frac{\partial \theta}{\partial p} - g \left(\frac{\partial u}{\partial p} \frac{\partial \theta}{\partial y} - \frac{\partial v}{\partial p} \frac{\partial \theta}{\partial x} \right), \quad (2.6)$$

in SI units $\text{K m}^2 \text{kg}^{-1} \text{s}^{-1}$ ($\equiv 10^{-6}$ PVU). The Wu Fortran evaluates an Exner-coordinate form of eq. (2.6); its exact relation to SI is derived in Chapter 4.

Interior PV exists only on the interior levels

`pvpialln` computes Q only on the *interior* levels $K = 2 \dots N_L - 1$ (850–200 hPa): the vertical derivative $\partial \theta / \partial \pi$ needs neighbours on both sides. At the surface ($K=1$, 1000 hPa) and top ($K=N_L$, 100 hPa) there is **no interior PV** — those two levels instead carry the *boundary potential temperature* of Section 2.4. This is why a surface-confined thermal anomaly shows “no PV at 1000 hPa” in the interior field.

2.4 Boundary Potential Temperature as a PV Sheet

The two boundary levels are represented by the half-level-averaged potential temperature (`pvpialln_94UV.f`, the lines that set `THB, THT`):

$$\theta_b^{(\text{bot})} = \frac{1}{2}(\theta_{K=1} + \theta_{K=2}), \quad \theta_b^{(\text{top})} = \frac{1}{2}(\theta_{K=N_L-1} + \theta_{K=N_L}). \quad (2.7)$$

In the `q.out` file the layout is therefore $[\theta_b^{(\text{bot})}, \theta_b^{(\text{top})}, Q_{K=2}, \dots, Q_{K=N_L-1}]$ — boundary θ first, interior PV after.

Through hydrostatic balance eq. (2.5), θ_b specifies the *vertical derivative* of the geopotential / streamfunction at the boundary, i.e. an **inhomogeneous Neumann boundary condition**. In `qinvert21_94.f` (BALNC) it enters the elliptic stencil one node inside each boundary (the lines that add $+\theta_b^{(\text{bot})}/\Delta\pi$ at $K=2$ and $-\theta_b^{(\text{top})}/\Delta\pi$ at $K=N_L-1$) and reconstructs the boundary layer by a half-step hydrostatic integration, $H_1 = H_2 + \theta_b^{(\text{bot})}(\pi_2 - \pi_1)$. If no boundary θ is supplied, the code applies *homogeneous* Neumann conditions.

Bretherton: surface theta as a boundary PV sheet

A warm surface potential-temperature anomaly is dynamically equivalent to a *positive* PV anomaly concentrated as a delta-function sheet at the lower boundary; a cold anomaly $\equiv$ a negative sheet. By the PV–inversion (“invertibility”) principle this induces a circulation:

$$\theta'_b > 0 \text{ (warm)} \Rightarrow \text{cyclonic } \psi' \text{ } (\psi' < 0 \text{ in NH}), \quad \theta'_b < 0 \text{ (cold)} \Rightarrow \text{anticyclonic.}$$

This is the textbook “PV-thinking” interpretation [2] of the Neumann condition above; the Fortran implements the boundary condition, not the interpretation. It is the reason the **surface piece is inverted separately** (its PV “lives in” the boundary θ), and why for this event the surface-piece ψ' is predominantly cyclonic (the domain-mean surface θ' is net warm, $\approx +1$ K).

2.5 Linearization and Balance Closure

Decompose $\psi = \bar{\psi} + \delta\psi$, $\Phi = \bar{\Phi} + \delta\Phi$. The piecewise solver inverts the PV/boundary- θ *perturbation* $q' = q_{\text{event}} - q_{\text{mean}}$ under a balance constraint. The production configuration uses the **nonlinear** balance (INLIN=1, Section 3.3): the operator coefficients are evaluated about the half-state $\bar{\cdot} + \frac{1}{2}(\cdot)'$ (CNLIN=0.5 in `qinvertp21_94.f`), so that the three piecewise solutions *sum to the total* perturbation,

$$\boxed{\delta\psi_{\text{surface}} + \delta\psi_{\text{lower}} + \delta\psi_{\text{upper}} \approx \delta\psi_{\text{total}}} \quad (2.8)$$

to solver tolerance (the linear-closure check of Section 5.6).

Why INLIN

Earlier notes claimed INLIN=0 (linear balance) was *mandatory* because the nonlinear balance “explodes at the jet.” On the current 9-level, extended-domain configuration the **nonlinear balance INLIN=1 converges** (all three pieces reach TOTAL CONVERGENCE) and is required for the additivity eq. (2.8). INLIN=0 is no longer used.

Chapter 3

The Wu Fortran Solver

This chapter describes the numerical core: W. Wu’s finite-difference piecewise PV-inversion code, driven from Python (`wu/steps/`) while the solver itself remains the original Fortran-77. The three programs live in `wu/fortran/` and are compiled with `gfortran -std=legacy -O2 -fno-automatic`.

3.1 Grid and Vertical Levels

The solver runs on an extended Northern-Hemisphere window (`config.py`, `wu/wu_config.yaml`):

Quantity	Symbol / code	Value
Longitudes	N_X	134
Latitudes	N_Y	51
Levels	$N_W = N_L$	9
Latitude row	$\phi_i = 85.5 - 1.5 i$	$85.5^\circ \rightarrow 10.5^\circ$ (N→S)
Longitude col	$\lambda_j = 120 + 1.5 j$	$120^\circ\text{E} \rightarrow 319.5^\circ\text{E}$ (= 40.5°W)
Pressure levels	PLEVS [hPa]	1000, 850, 700, 500, 400, 300, 250, 200, 100
Earth radius	$a = 2 \times 10^7 / \pi$	6.366×10^6 m
Grid spacing	$\Delta p = \Delta l = a \text{ rad}(1.5^\circ)$	1.667×10^5 m

Fortran PARAMETER blocks are hard-wired

N_X, N_Y, N_W appear as compile-time **PARAMETERS** inside *every* `.f` file (and inside each subroutine). Changing the grid requires editing and recompiling the Fortran — not just `config.py`. The current 9-level build sets `NW=9`, `NL=9`, and a 9-value `DATA PR` array in all three sources. The Python drivers pass solver *parameters* (relaxation factors, piece definitions) over `stdin`; they cannot change array dimensions.

3.2 The Three Programs and Four Passes

Pass	Program	Subroutine	Produces
A	<code>pvpialln</code>	—	<code>meanq</code> , <code>meanh</code> (mean Q , θ_b , ψ)
B	<code>pvpialln</code>	—	<code>event_q.out</code> , <code>event_h.out</code>
C	<code>qinvert21</code>	BALNC	<code>event_bal.out</code> (total balanced ψ , Φ)
D	<code>qinvert21</code>	BALP	<code>event_pert.out</code> (3-piece ψ')

Pass A/B — `pvpialln`. Computes the boundary potential temperatures $\theta_b^{\text{bot}}, \theta_b^{\text{top}}$ (Section 2.4), the Ertel-like PV on the interior levels ($K = 2 \dots N_W - 1$, i.e. 850–200 hPa), the balanced streamfunction ψ from relative-vorticity inversion, and the geopotential. Pass A runs on the 30-year climatological mean state; Pass B on the 2025-01-08 event state. The `q.out` layout is $[\theta_b^{\text{bot}}, \theta_b^{\text{top}}, Q(K=2..N_W-1)]$.

Pass C — `qinvert21 (BALNC)`. The full *total* balanced inversion: a 2-D Helmholtz problem for ψ on each level coupled to a 3-D PV– ψ relation under gradient-wind balance, by SOR with under-relaxation between ψ and Φ . The boundary θ enters as the inhomogeneous Neumann condition of Section 2.4.

Pass D — `qinvertp21 (BALP)`. Partitions the PV *perturbation* $q' = q_{\text{event}} - q_{\text{mean}}$ into three pieces and inverts each independently to its perturbation streamfunction ψ' . In the BALP convention a piece is a list QLV of level indices, where the value 1 activates the *lower-boundary* θ , NL the *upper-boundary* θ , and $2..N_L-1$ the interior PV at that level:

Piece	QLV	Activates	Pressures
surface	{1}	1000 hPa boundary θ only	—
lower	{2, 3}	interior PV	850, 700 hPa
upper	{4, 5, 6, 7, 8, 9}	interior PV + top θ ($9=N_L$)	500–200 hPa + 100 θ

The three QLV lists cover every field (θ_b^{bot} , interior PV at $K=2..8$, θ_b^{top}) exactly once, which is the precondition for the additivity eq. (2.8). The stdin actually sent is `1,1 / 2,2,3 / 5,4,5,6,7,8,9`.

The surface piece is the surface heating

Because QLV={1} activates only the lower-boundary θ , the **surface piece is the inversion of the surface warm/cold- θ anomaly** (the Bretherton PV sheet, Section 2.4) — not of any interior 1000 hPa PV (there is none). This isolates the surface thermal forcing from the free-tropospheric PV, mirroring the reference `inv3d` partition.

3.3 Solver Parameters

Parameters are passed over `stdin` from `wu_config.yaml`:

Param	Pass C	Pass D	Role
OMEGS	1.4	1.85	SOR factor, ψ sweep
OMEGH	1.4	1.4	SOR factor, Φ/H sweep
PART	0.05	0.05	under-relaxation $\psi \leftrightarrow \Phi$
THRSH	0.01	0.1	convergence threshold
INLIN	—	1	nonlinear balance (additivity)
IBC	—	0	homogeneous Dirichlet lateral walls

`qinvertp21_94.f` *hardcodes* its iteration caps and PV floor (`MAX = 8000`, `MAXT = 2000`, `QMIN = 10-5`); the YAML does not set them. The non-dimensional Exner increment is `DPI = 500/ $N_L$  = 55.56` (auto-scaled with N_L).

3.4 Convergence and Closure (current behaviour)

- **Pass D** converges cleanly: all three pieces report `TOTAL CONVERGENCE`, no NaNs, with physical magnitudes (the upper piece dominates the 250 hPa response).
- **Pass C** prints `PHI CONVERGED` (per-cycle residual $\sim 10^{-4}$, below threshold) and then a *benign* “started diverging” from `qinvert21`’s outer-coupling heuristic (it fires when an outer cycle needs > 10 more inner iterations after > 30 cycles). The written `event_bal.out` is the per-cycle-converged field (identical at $\omega = 1.2$ and 1.4); it is used as the total reference for Pass D.
- **Top boundary 100 hPa is non-uniform** ($\Delta\sigma$ jumps $0.05 \rightarrow 0.10$ from 200 to 100 hPa). This is accepted: the 9-level grid closes the piecewise sum substantially better (linear-closure $R^2 \approx 0.44$ at 250 hPa) than a uniform-top 8-level grid ($R^2 \approx -0.2$), and all analysis of interest is at ≤ 200 hPa.

Chapter 4

Wu PV to SI Ertel PV: Exact Conversion

This chapter documents the *exact analytic* conversion from the Wu Fortran PV scalar Q_{Wu} to true SI Ertel potential vorticity. The conversion coefficient is **exactly** 10^{-8} at every pressure level when $\kappa = 2/7$, which is the standard dry-air value used throughout the Wu code. This result was derived on 2026-06-03 and replaces an earlier ad-hoc median-ratio scaling.

4.1 Wu PV Definition

In `pvpialln_94UV.f` (lines 460–468), Ertel-like PV is computed as

$$Q_{\text{Wu}} = -\text{COEF} \cdot \Pi^{-2.5} \left[(f + \zeta) \frac{\partial \theta}{\partial \Pi} + \left(\frac{\partial u}{\partial \Pi} \frac{\partial \theta}{\partial y} - \frac{\partial v}{\partial \Pi} \frac{\partial \theta}{\partial x} \right) \right], \quad (4.1)$$

where

$$\text{COEF} = \frac{10^2 \cdot 10^6 \cdot g \cdot \kappa \cdot c_p^{3.5}}{p_0}, \quad (4.2)$$

$$\Pi(p) = c_p \left(\frac{p}{p_0} \right)^\kappa, \quad (4.3)$$

with numerical constants $g = 9.81 \text{ m s}^{-2}$, $\kappa = 2/7$, $c_p = 1004.5 \text{ J kg}^{-1} \text{ K}^{-1}$, $p_0 = 10^5 \text{ Pa}$ (but see warning below). The “1.E2” in the COEF expression is a legacy over-scaling of no physical significance; it is the conjugate of the 10^{-2} divide applied to PV in the non-dimensional BALNC/BALP solvers, and it cancels exactly in the back-conversion derived below.

4.2 True SI Ertel PV

The standard definition of Ertel PV in isobaric coordinates is

$$\text{PV}_{\text{SI}} = -g \left[(f + \zeta) \frac{\partial \theta}{\partial p} + \frac{\partial u}{\partial p} \frac{\partial \theta}{\partial y} - \frac{\partial v}{\partial p} \frac{\partial \theta}{\partial x} \right], \quad (4.4)$$

with units $\text{K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$ ($\equiv 10^{-6} \text{ PVU}$).

4.3 Conversion

The two definitions, (4.1) and (4.4), differ only in the vertical coordinate of differentiation: $\partial/\partial\Pi$ in Wu versus $\partial/\partial p$ in SI. The chain rule relates them:

$$\frac{\partial}{\partial p} = \frac{\partial\Pi}{\partial p} \frac{\partial}{\partial\Pi} = \frac{\kappa\Pi}{p} \frac{\partial}{\partial\Pi}. \quad (4.5)$$

The factor $\kappa\Pi/p$ multiplies *both* the vorticity-stability term and the shear term identically, so it factors out of the entire bracket:

$$\begin{aligned} \text{PV}_{\text{SI}} &= Q_{\text{Wu}} \cdot \frac{g}{\text{COEF}} \cdot \frac{\kappa\Pi}{p} \cdot \Pi^{2.5} \\ &= Q_{\text{Wu}} \cdot \frac{g\kappa}{\text{COEF}} \cdot \frac{\Pi^{3.5}}{p} \equiv Q_{\text{Wu}} \cdot c(p). \end{aligned} \quad (4.6)$$

4.3.1 The Coefficient $c(p)$ is Constant

Substituting $\Pi = c_p(p/p_0)^\kappa$ and $\text{COEF} = 10^8 g \kappa c_p^{3.5}/p_0$:

$$\begin{aligned} c(p) &= \frac{g\kappa}{10^8 g \kappa c_p^{3.5}/p_0} \cdot \frac{c_p^{3.5}(p/p_0)^{3.5\kappa}}{p} \\ &= \frac{1}{10^8} (p/p_0)^{3.5\kappa-1}. \end{aligned} \quad (4.7)$$

With $\kappa = 2/7$,

$$3.5\kappa = \frac{7}{2} \cdot \frac{2}{7} = 1 \quad \implies \quad c(p) = \frac{1}{10^8} (p/p_0)^0 = \mathbf{10^{-8}} \quad (4.8)$$

at *every* pressure level. Wu’s choice of the $\Pi^{-2.5}$ weighting was designed precisely so that $\Pi^{3.5}/p$ is pressure-independent: the p dependence in $\Pi^{3.5}$ exactly cancels with the $1/p$ from the chain rule.

Wu $Q \rightarrow$ SI conversion is a universal constant

$$\boxed{\text{PV}_{\text{SI}} = Q_{\text{Wu}} \times 10^{-8}} \quad \text{or} \quad \boxed{Q_{\text{Wu}} = \text{PV}_{\text{SI}} \times 10^8}$$

This holds for *any* pressure level when $\kappa = 2/7$. If a different κ were used, the per-level array $c(p) = (g\kappa/\text{COEF}) \cdot \Pi^{3.5}/p$ must be evaluated; the ‘WU2SI’ array in `parse_and_pv.py` implements this general form.

4.4 Agreement with ERA5 (after the raw- T fix)

Dividing Wu Q by 10^8 converts it to *genuine SI units* ($\text{K m}^2 \text{kg}^{-1} \text{s}^{-1}$). Once the input bug was fixed — the grid writer now feeds `pvpialln raw temperature  $T$` , not θ , so the Exner conversion is applied exactly once — the converted field agrees with ERA5’s native Ertel PV essentially one to one. At 250 hPa the current pipeline reports

$$\text{median ratio } \frac{Q_{\text{Wu}} \times 10^{-8}}{\text{PV}_{\text{ERA5}}} = 1.00, \quad \text{slope} = 0.99, \quad \text{corr} = 0.995. \quad (4.9)$$

History: the spurious 3.5x residual

An earlier version fed θ into `vpialln`, which then applied the Exner factor a *second* time. This double conversion inflated the static stability $\partial\theta/\partial\Pi$ and hence Q by a level-dependent $\sim 2\text{--}4\times$ (median $\sim 3.5\times$), and was masked at the time by an ad-hoc `STAB_SCALE` ≈ 0.3 . Both the double conversion and the fudge factor are gone; the residual factor is now ≈ 1 . No empirical scaling is applied — only the analytic 10^{-8} .

(Equivalently, since $\text{PVU} = 10^{-6} \text{ SI}$, the Wu scalar is $\approx 100\times$ a PVU value: $\text{PVU} = Q_{\text{Wu}} \times 10^{-2}$.)

4.5 Implementation in the Pipeline

In `wu/steps/08_parse_outputs/parse_and_pv.py`:

```
1 # Exact Wu Q -> SI Ertel PV per-level coefficient
2 PLEVS_PA = np.asarray(WU_LEVS) * 100.0
3 PI_LEV   = CP * (PLEVS_PA / P0) ** KAP
4 COEF_WU  = 1.0e2 * 1.0e6 * 9.81 * KAP * CP**3.5 / P0
5 WU2SI    = (G * KAP / COEF_WU) * PI_LEV**3.5 / PLEVS_PA # ~1e-8
6 Q_wu_si  = Q_wu * WU2SI[k250] # SI [K.m2.kg-1.s-1]
7 Q_wu_pvu = Q_wu_si * 1e6 # PVU for display
```

The $c(p) = 10^{-8}$ constant is also documented in `config.py` section 7 alongside the full Wu non-dimensional scale set (DPI, LL, FF, THO, HND, QCONST, PI_WU, PIF, BB/BH/BL).

COEF uses Fortran g

The Wu Fortran hardcodes $g = 9.81$ (line 460), while the standard SI value is $g = 9.80665 \text{ m s}^{-2}$. The Python coefficients match the Fortran's 9.81 for exact numeric consistency. The difference is $< 0.04\%$ and is absorbed by the residual discretisation bias anyway.

Chapter 5

The Wu Pipeline, Step by Step

This chapter walks through the executable steps in `wu/steps/` that take the raw Fortran output to the piecewise diagnostics. Each step is illustrated with the figure it generates (regenerated for this handbook into `handbook/figs/wu/`).

5.1 Step 05 — Pass A/B: PV, Boundary θ , Balanced ψ

Step 05 compiles the three Fortran programs and runs `pvpialln` twice: Pass A on the 30-year mean state (`meanq`, `meanh`) and Pass B on the 2025-01-08 event state (`event_q.out`, `event_h.out`). It computes the boundary θ (θ_b^{bot} at 1000, θ_b^{top} at 100 hPa), the interior Ertel PV on 850–200 hPa, and the balanced streamfunction.

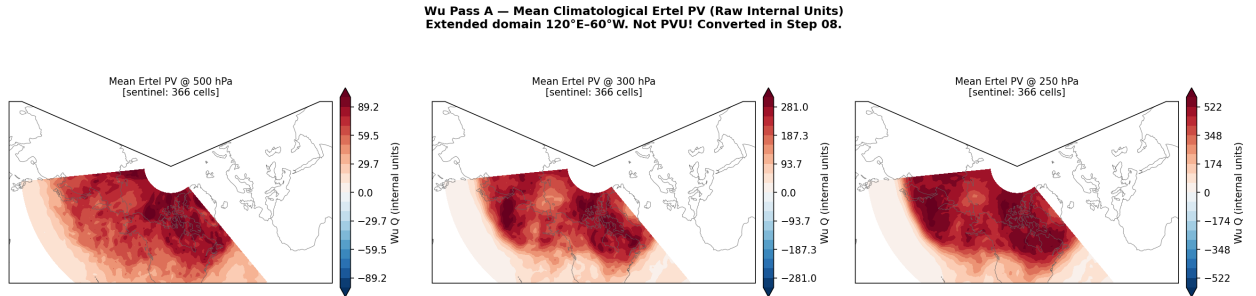


Figure 5.1: **Step 05 / Pass A.** Mean-state Ertel PV (Wu internal units) at three representative *interior* levels. The upper-level panel shows the climatological jet PV gradient that the event anomaly perturbs.

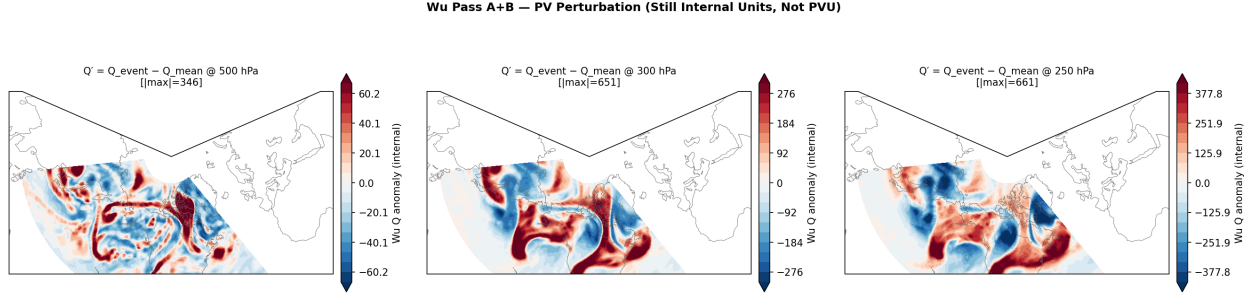


Figure 5.2: **Step 05 / Pass B.** Event-minus-mean PV anomaly q' . The dominant upper-tropospheric positive anomaly is the blocking ridge inverted in the *upper* piece of Pass D; the weak low-level anomaly is dwarfed by it (and the surface heating itself lives in the boundary θ , not in this interior field).

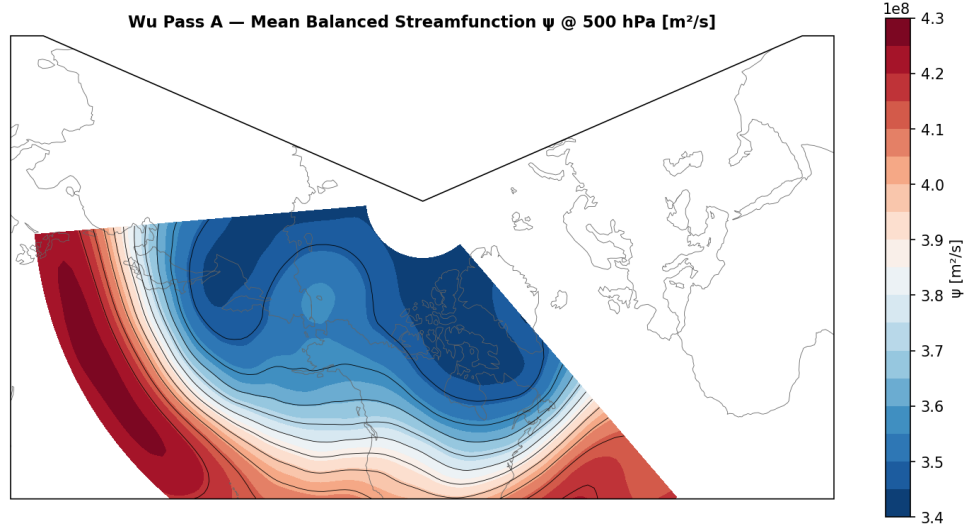


Figure 5.3: **Step 05.** Balanced mean-state streamfunction $\bar{\psi}$ at 500 hPa — the basic state about which the nonlinear (INLIN=1) perturbation inversion is linearised.

5.2 Step 06 — Pass C: Total Balanced Inversion (BALNC)

Step 06 runs `qinvert21` (BALNC) to produce the total balanced (ψ, Φ) pair (`event_bal.out`), the reference from which Pass D forms the perturbation. Per-cycle convergence is reached ($\sim 10^{-4}$, below threshold); the trailing “started diverging” message is the benign outer-coupling heuristic of Section 3.4, and the written field is the converged state.

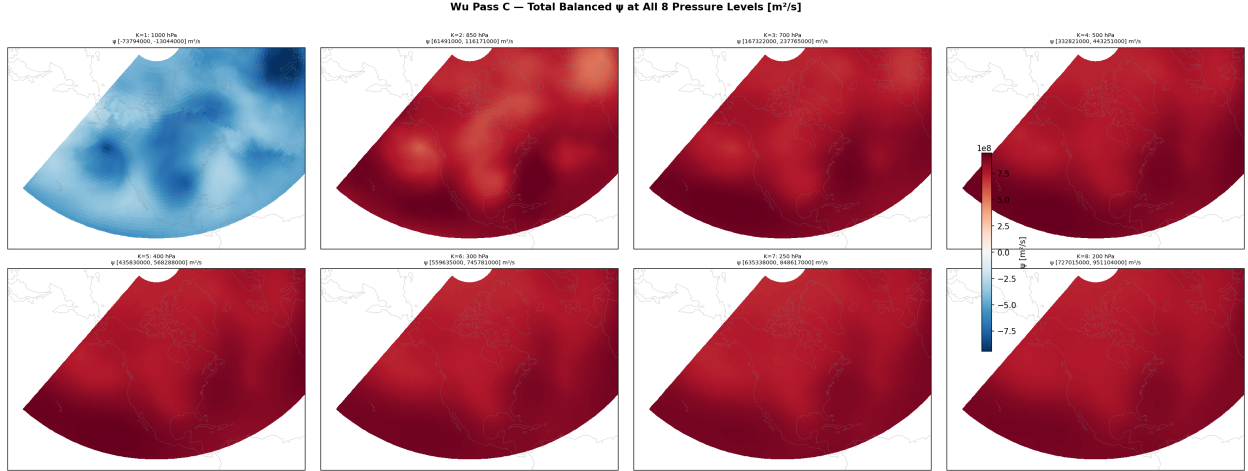


Figure 5.4: **Step 06 / Pass C.** Total balanced streamfunction at all nine pressure levels, showing the equivalent-barotropic vertical structure of the blocking circulation.

5.3 Step 07 — Pass D: Piecewise Perturbation Inversion (BALP)

Step 07 runs `qinvertp21` (BALP), partitioning q' into the **surface**, **lower** and **upper** pieces (Section 3.2) and inverting each to its perturbation streamfunction ψ' . All three report **TOTAL CONVERGENCE**.

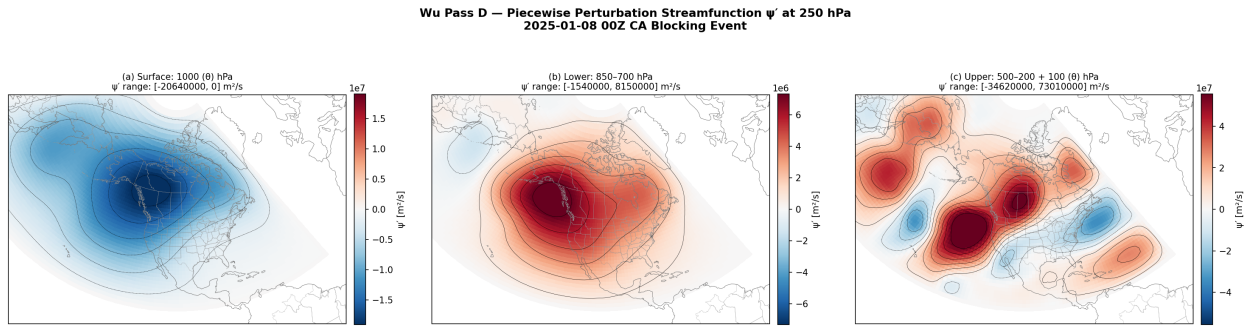


Figure 5.5: **Step 07 / Pass D.** 250 hPa perturbation streamfunction ψ' induced by each piece: (a) surface (1000 hPa boundary θ), (b) lower (850–700 hPa PV), (c) upper (500–200 hPa PV + 100 hPa top θ). The upper piece dominates the jet-level response; the surface piece is a broad, predominantly cyclonic ($\psi' < 0$) response to the net-warm surface θ' .

5.4 Step 08 — Wu Q vs ERA5 Ertel PV

Step 08 parses all Fortran outputs, converts Wu Q to SI by the exact 10^{-8} constant (Chapter 4), and computes the induced winds and PV advection, writing `piecewise_psi.nc` and `pv_advection.nc` (all SI).

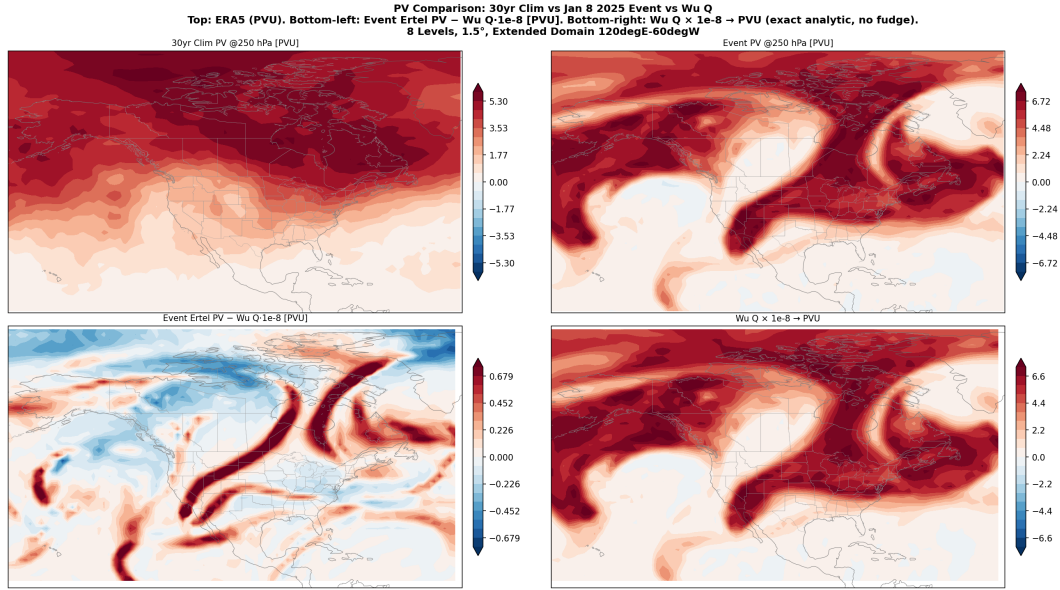
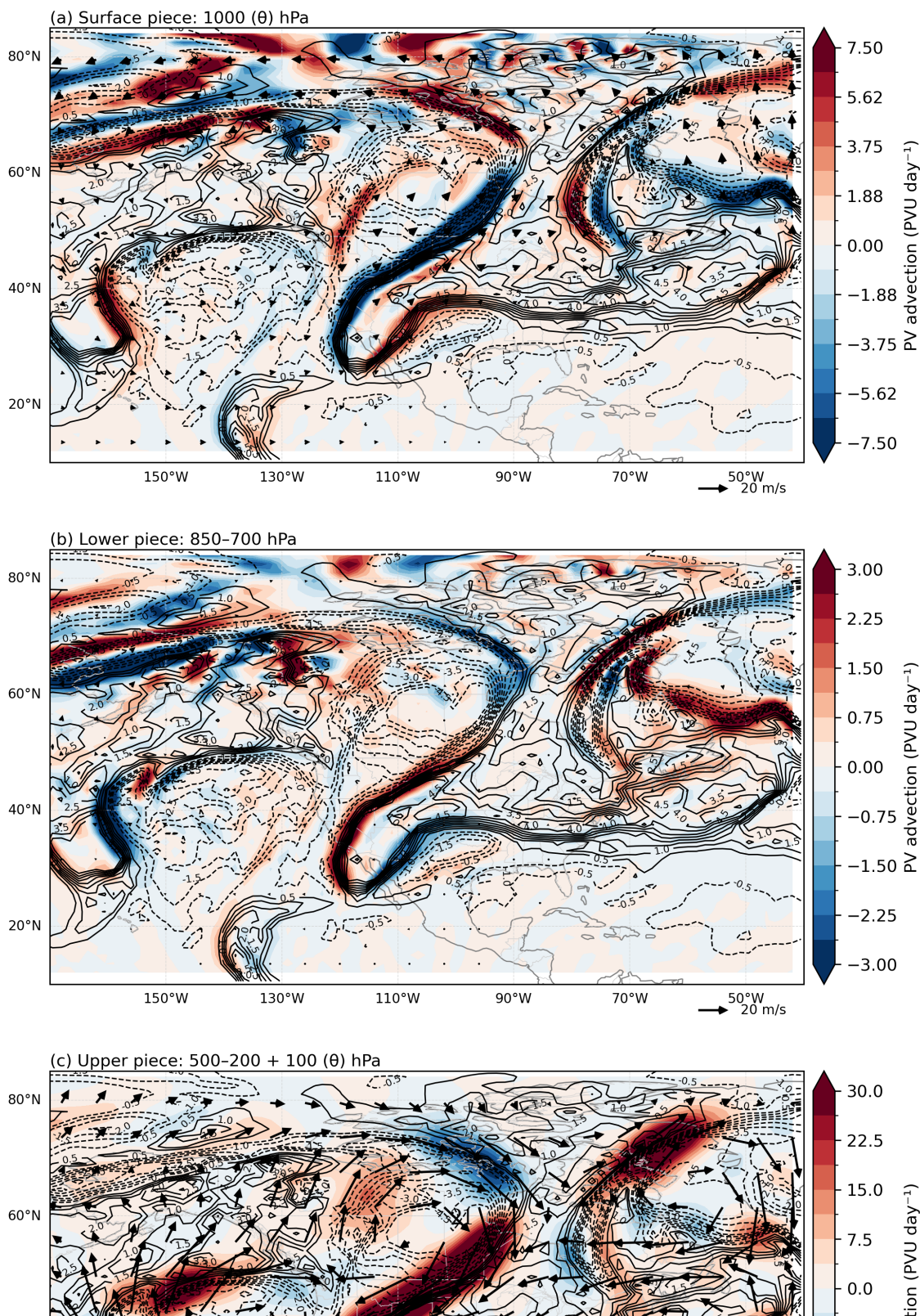


Figure 5.6: **Step 08.** Wu $Q \times 10^{-8}$ versus ERA5 native Ertel PV at 250 hPa. After the raw- T fix the median ratio is 1.00 (corr = 0.995); no empirical scaling is used.

5.5 Step 10 — Fig. 8 Replica (PV Advection)

250-hPa PV Advection Induced by Piecewise PV Anomalies
2025-01-08 00Z | INLIN=1 | 9 Levels | 30yr Clim



5.6 Step 11 — Cross-Sections and 3-D Structure

Step 11 produces the latitude/longitude–pressure cross-sections and a 3-D “vertical structure” view, plus the linear-closure check.

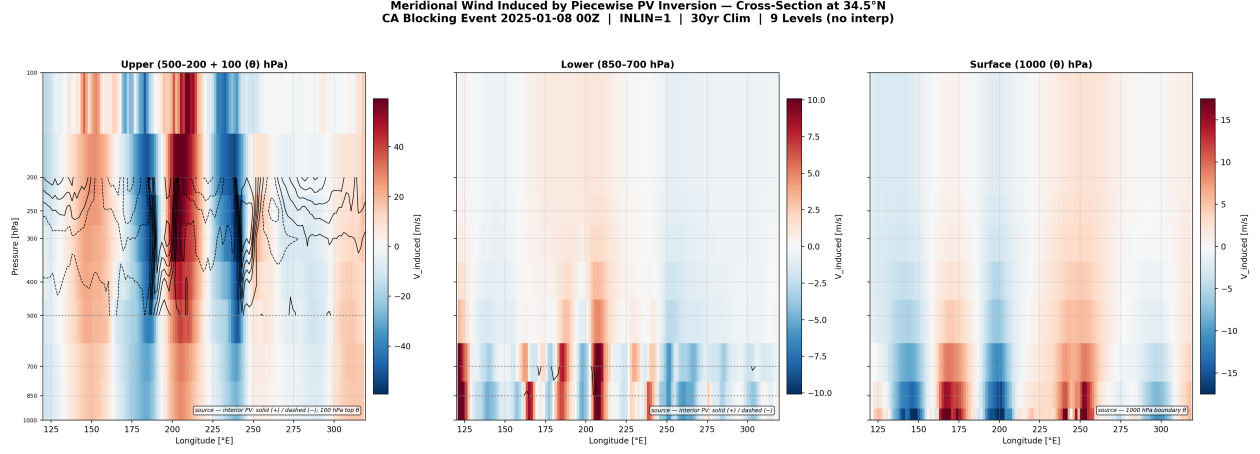


Figure 5.8: **Step 11.** Zonal–vertical cross-section at 34.5°N of the meridional wind v' induced by each piece, with each piece’s *own* source overlaid (interior PV contours; boundary- θ pieces labelled). The **upper** piece peaks aloft (~ 250 hPa, $\pm 50 \text{ m s}^{-1}$); the **surface** piece is surface-trapped (peaks at 1000 hPa, decaying upward) — exactly the Bretherton boundary response.

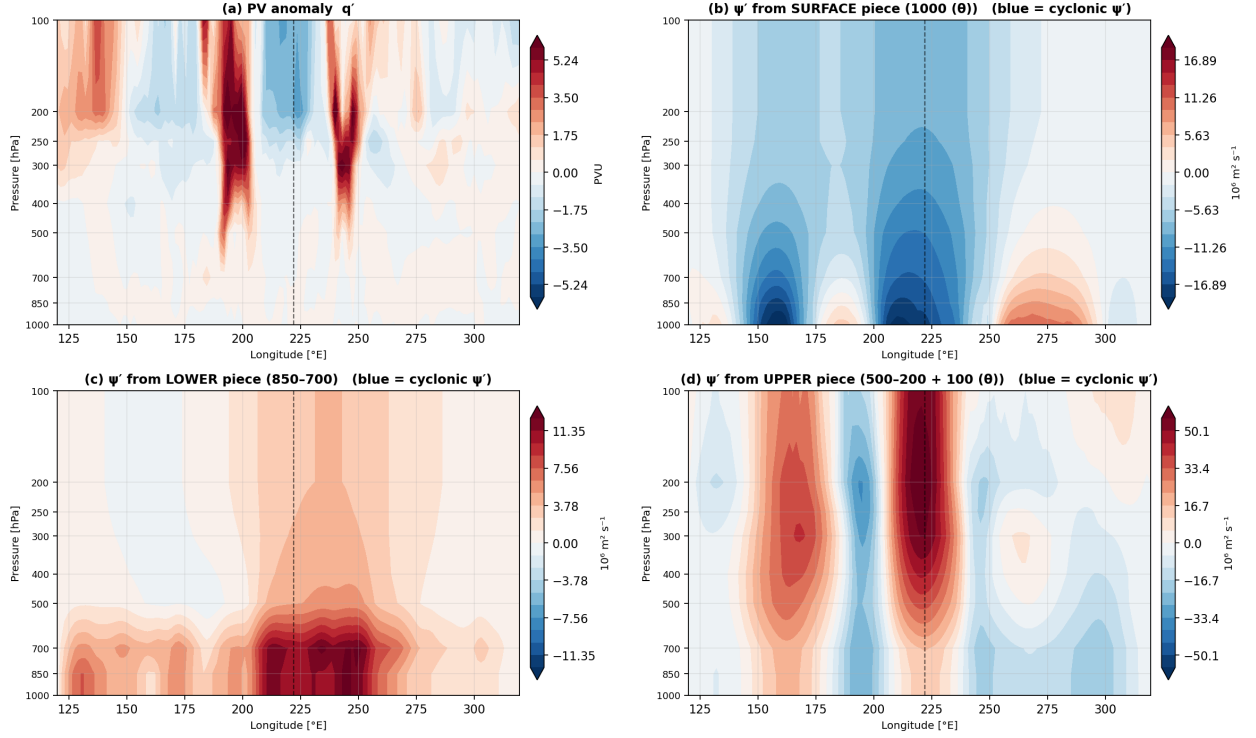


Figure 5.9: **Step 11.** Longitude–pressure (x – p) cross-sections at the block latitude (31.5°N; dashed line = block longitude 222°E): **(a)** total PV anomaly q' ; **(b–d)** ψ' induced by the surface, lower and upper pieces (blue = cyclonic $\psi' < 0$). The PV anomaly is concentrated at upper levels (panel a); the surface boundary level carries no interior PV — the surface heating is the boundary θ inverted in panel (b), a broad predominantly cyclonic response. The upper piece (d) dominates the amplitude.

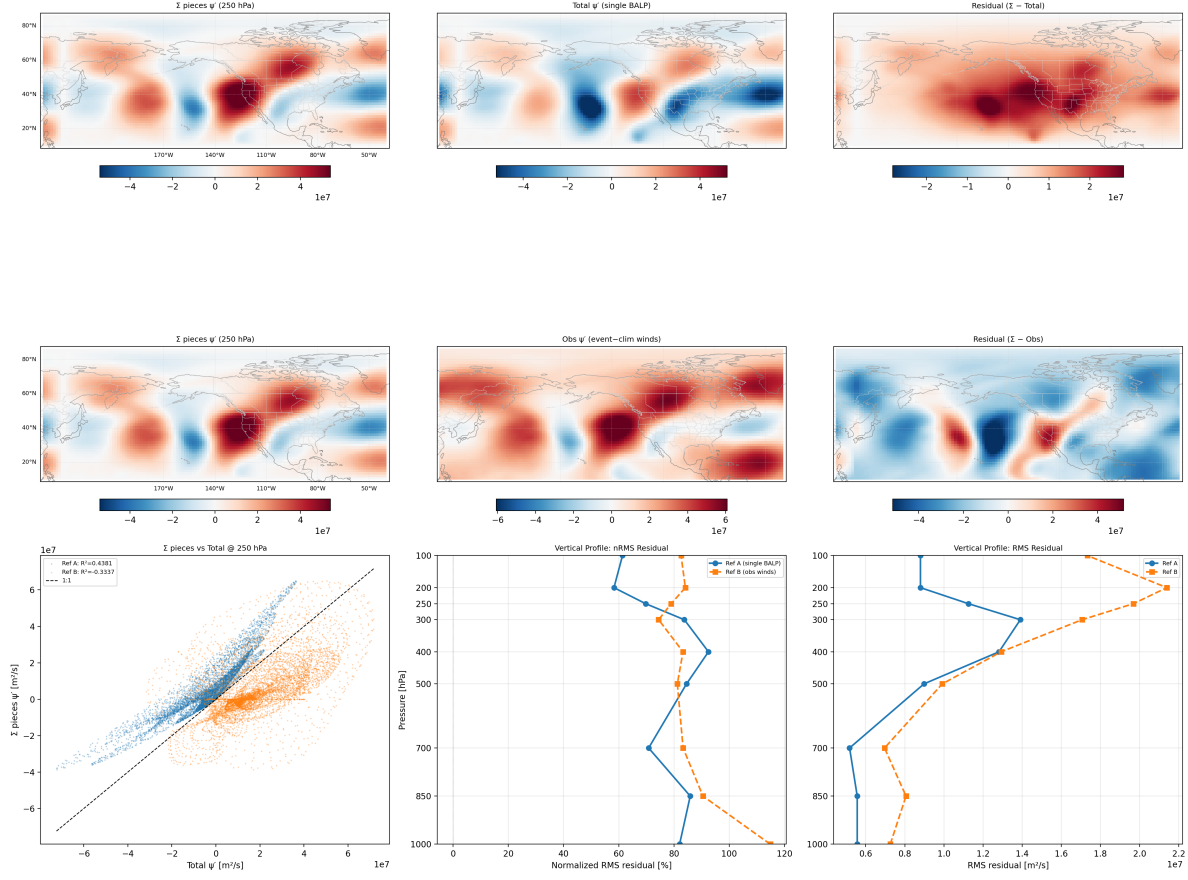


Figure 5.10: **Step 11.** Linear-closure check: the sum of the three piecewise ψ' against a single all-levels inversion. On the 9-level (top-100) grid the agreement is moderate ($R^2 \approx 0.44$ at 250 hPa) — a real grid effect, not a solver error (Section 3.4).

Chapter 6

Conclusions

6.1 Summary

This handbook documents the **Wu Fortran piecewise PV inversion pipeline** [1] applied to the 2025-01-08 California blocking event, on a nine-level ERA5 grid with the perturbation split into a **surface**, **lower** and **upper** piece:

1. **Download / climatology / grid** (`shared_steps/01-04`) — ERA5 event hour, 30-year mean, and the Fortran `.grid` inputs.
2. **Pass A/B** (`pvpialln`) — boundary θ , interior Ertel PV (850–200 hPa), balanced ψ , for the mean and event states (step 05).
3. **Pass C** (`qinvert21/BALNC`) — total balanced reference (step 06).
4. **Pass D** (`qinvertp21/BALP`) — the three-piece perturbation inversion, `INLIN=1` so the pieces sum to the total (step 07).
5. **Diagnostics** — Wu-vs-ERA5 PV, induced winds, PV advection, cross-sections, 3-D structure, closure (steps 08–11).

6.2 Key Findings

1. **Surface heating is a boundary condition, not interior PV.** Interior Ertel PV exists only at $K=2..N_L-1$ (850–200 hPa); the surface (1000) and top (100) levels carry the boundary θ . By Bretherton’s equivalence the warm surface θ' is a positive PV *sheet* at the boundary, applied as an inhomogeneous Neumann condition. We invert it as its own **surface piece** (`QLV={1}`), isolating the surface thermal forcing.
2. **The surface-piece ψ' is predominantly cyclonic for this event** ($\sim 84\%$ of points have $\psi' < 0$ at 1000 hPa), because the domain-mean surface θ' is net warm ($\approx +1$ K). This is case-dependent, not a law: a net-cold surface anomaly would give an anticyclonic surface piece.
3. **Wu $Q \rightarrow$ SI is the exact constant 10^{-8} when $\kappa = 2/7$** (Chapter 4). After the raw- T input fix, $Q_{\text{Wu}} \times 10^{-8}$ matches ERA5 native Ertel PV with median ratio 1.00, `corr = 0.995`. The former “ $\sim 3.5 \times$ residual” and any empirical scaling are retired.

4. **The 9-level (top-100) grid closes the piecewise sum better** ($R^2 \approx 0.44$ at 250 hPa) than a uniform-top 8-level (top-200) grid ($R^2 \approx -0.2$). The non-uniform top is accepted because all analysis of interest is at ≤ 200 hPa.
5. **Pass D converges (INLIN=1)**; the earlier “INLIN=0 mandatory” claim is obsolete. Pass C’s “started diverging” is a benign post-convergence heuristic.

Table 6.1: SI unit conventions for all pipeline outputs.

Quantity	Symbol	Units
Streamfunction	ψ	$\text{m}^2 \text{s}^{-1}$
Geopotential	$\Phi = gz$	$\text{m}^2 \text{s}^{-2}$
Ertel PV	q	$\text{K m}^2 \text{kg}^{-1} \text{s}^{-1}$
PV advection tendency	$-\mathbf{u} \cdot \nabla q$	$\text{K m}^2 \text{kg}^{-1} \text{s}^{-2}$
Induced wind	u, v	m s^{-1}
Pressure	p	hPa

6.3 Limitations and Future Work

- **Flat lower boundary.** The solver treats 1000 hPa as a rigid lid everywhere and ignores topography; over the western-US land surface the true lower boundary is well above 1000 hPa. A terrain-following / surface-pressure lower boundary ($p = \text{hyam } p_0 + \text{hybm } p_s$) with the actual near-surface θ would be the principal physical upgrade. The boundary- θ Neumann condition itself is standard and correct; the GFDL/CSU PPVI tool uses the identical scheme (plus a regional “surgery” mask of the boundary θ).
- **Coarse vertical/horizontal resolution** (1.5° , 9 levels) limits the static-stability estimate and the closure.

Appendix A

Notation Glossary

Symbol	Units	Description
a	m	Earth radius 6.371×10^6
α	–	Picard under-relaxation parameter
$\beta = 2\Omega \cos \phi/a$	$\text{s}^{-1} \text{m}^{-1}$	meridional grad. of f
$\bar{\Phi}$	m^2/s^2	climatological mean geopotential
$\bar{\psi}$	m^2/s	climatological mean streamfunction
$\bar{q}$	PVU	climatological mean Ertel PV
$\bar{\zeta} = \nabla^2 \bar{\psi}$	s^{-1}	mean relative vorticity
c_p	J/kg/K	specific heat at const. pressure (1004)
$\delta\Phi = \Phi - \bar{\Phi}$	m^2/s^2	geopotential anomaly
$\delta\psi = \psi - \bar{\psi}$	m^2/s	streamfunction anomaly
$q' = q - \bar{q}$	PVU	Ertel PV anomaly
D_1, D_2	–	1st/2nd derivative matrices in π
$f = 2\Omega \sin \phi$	s^{-1}	Coriolis parameter
$f_0 = f(45^\circ)$	s^{-1}	reference Coriolis
g	m/s^2	gravity (Fortran 9.81)
$J(a, b)$	varies	Jacobian
$\kappa = R_d/c_p$	–	$2/7 \approx 0.2857$ (Wu hardcoded)
∇^2, ∇^{-2}	m^{-2}	horizontal Laplacian and its inverse
θ_b	K	boundary potential temperature (surface/top)
Ω	rad/s	Earth rotation rate 7.292×10^{-5}
$\pi = (p/p_0)^\kappa$	–	Exner pressure
p_0	Pa	reference pressure 10^5
ϕ, λ	rad	latitude, longitude
ψ, Φ	$\text{m}^2/\text{s}, \text{m}^2/\text{s}^2$	streamfunction, geopotential
PVU	$10^{-6} \text{K m}^2/\text{kg/s}$	potential vorticity unit
Q, q	PVU	Ertel potential vorticity
$\theta = T(p_0/p)^\kappa$	K	potential temperature
$\zeta = \nabla^2 \psi$	s^{-1}	relative vorticity

Appendix B

Public-API Reference

B.1 Configuration

B.1.1 `config.py` and `wu/wu_config.yaml`

- Grid: $N_X=134$, $N_Y=51$, $N_W=9$; $PLEVS = [1000, 850, 700, 500, 400, 300, 250, 200, 100]$ hPa. Domain $85.5^\circ\text{N} \rightarrow 10.5^\circ\text{N}$, $120^\circ\text{E} \rightarrow 319.5^\circ\text{E}$.
- PIECES: surface $\{1\}$, lower $\{2, 3\}$, upper $\{4, 5, 6, 7, 8, 9\}$ (mirrored authoritatively in `wu_config.yaml` pieces).
- Solver params live in `wu_config.yaml` (`pass_ab`, `pass_c`, `pass_d`); `config.py` keeps grid, physical constants, the exact 10^{-8} Wu $\rightarrow$ SI note, and paths.
- Paths: `ERA5_DIR`, `CLIM_DIR`, `WU_IN_DIR`, `WU_OUT_DIR`, `FIG_DIR`.

B.2 Fortran programs (`wu/fortran/`)

- `pvpialln_94UV.f` — Pass A/B: boundary θ (THB,THT), interior Ertel PV, balanced ψ, Φ (Section 3.2). `PARAMETER (NW=9)`.
- `qinvert21_94.f` (BALNC) — Pass C: total balanced inversion. `PARAMETER (NL=9)` in main and BALNC.
- `qinvertp21_94.f` (BALP) — Pass D: 3-piece perturbation inversion; QLV selects boundary θ ($1 / NL$) and interior PV ($2..N_L-1$). `PARAMETER (NL=9)` in main and BALP; hardcodes `MAX=8000`, `MAXT=2000`, `QMIN=1e-5`, `DPI=500/NL`.

B.3 Shared front end (`shared_steps/`)

- `01_download/download_era5.py` — event hour at the 9 levels (derived from `config.PLEVS`).
- `03_climatology/clim_30yr_mean.py` — 30-yr mean + event subset to the 9 Wu levels (`mean_clim.nc`, `event.nc`).
- `04_write_grid/write_grid_files.py` — writes `mean.grid`, `event.grid` ($4N_W$ field-blocks).

B.4 Wu step drivers (`wu/steps/`)

- `05_wu_pass_ab/` — compile + Pass A/B; writes `meanq`, `meanh`, `event_{q,h}.out`.

- 06_wu_pass_c/ — Pass C; writes `event_bal.out`.
- 07_wu_pass_d/ — Pass D; writes `event_pert.out` (3 pieces, QLV from `wu_config.yaml`).
- 08_parse_outputs/parse_and_pv.py — Wu $Q \rightarrow$ SI ($\times 10^{-8}$); writes `piecewise_psi.nc`, `pv_advection.nc` (all SI).
- 10_fig8/fig8_replica.py — Davis-style PV-advection diagnostic (surface/lower/upper).
- 11_xsection/ — `xsection_vinduced.py` (zonal cross-section), `xsection_3d_perpendicular.py` (3-D vertical structure, Fig.-16 style), `upper_psi_by_piece.py`, `closure_check.py`, `closure_diagnostic.py`.

B.5 Key output: `piecewise_psi.nc`

- Dims (piece, plev, lat, lon) = (3, 9, 51, 134); the piece axis is ordered `surface`, `lower`, `upper`.
- Variables `psi`, `geopotential`, `U_induced`, `V_induced`, all SI.

Bibliography

- [1] C. A. Davis and K. A. Emanuel. Potential vorticity diagnostics of cyclogenesis. *Monthly Weather Review*, 119:1929–1953, 1991.
- [2] B. J. Hoskins, M. E. McIntyre, and A. W. Robertson. On the use and significance of isentropic potential vorticity maps. *Quarterly Journal of the Royal Meteorological Society*, 111(470):877–946, 1985.